

INFOSYS SPRINGBOARD INTERNSHIP — MODULE 4

Unified Military Analytics Dashboard

Dashboard Planning, Storyboard & Interactive Prototype

PREPARED BY

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BUILT WITH

Python · Streamlit · Pandas · Plotly ·
OpenPyXL

DELIVERABLE TYPE

Storyboard + Functional Prototype

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This document consolidates the dashboard planning artefacts for **Module 4** of the Infosys Springboard Internship. It presents the information architecture, the interaction design decisions, and the working prototype screens captured from the deployed **Streamlit** application, which visualises global military capability data across active personnel, assets, budgets and force composition for 190+ nations.

The **Unified Military Analytics Dashboard** is an interactive Streamlit application developed to explore and compare national military capability using a cleaned, curated military dataset. The prototype demonstrates the full dashboard planning lifecycle — from wireframing and navigation design to a functioning, data-driven interface — and forms the Module 4 deliverable of the Infosys Springboard Internship programme.

The application allows a user to select any country from a searchable filter and instantly view its key military indicators, asset composition, and a full, sourced breakdown of every underlying metric — from total population and available manpower through to naval tonnage and fighter fleet strength.

Core Capabilities

- **Quick Stats** — headline KPIs: Military Power Index, Global Rank, Defense Budget and Active Personnel at a glance.
- **Nation Overview** — a country-level deep dive combining KPI cards, asset visualisations and summary statistics.
- **Compare Powers** — side-by-side visual comparison of military strength between two or more nations.
- **Coalition Builder** — a prototype workspace for aggregating the combined capability of a selected group of countries.

Status: The *Nation Overview* module is fully operational in the current build, satisfying the Module 4 requirement that at least one dashboard (Quick Stats or Nation Overview) be functional end-to-end with live filtering, KPI computation and chart rendering.

Dashboard Architecture & Navigation Logic

Navigation follows a single-page, sidebar-driven pattern typical of Streamlit multi-view apps: a persistent left-hand **Filters** panel controls global state (the selected country), while the main canvas swaps between stacked content modules that the user scrolls through — Key Metrics, Military Assets, and the full sourced Country Data table.

Country Filter	Sidebar dropdown / search	Re-queries the dataset and refreshes every downstream card, chart and table.
Key Metrics	Auto (on filter change)	Recomputes Active Personnel, Assets-per-Capita and Budget-to-GDP ratio.
Military Asset Strength	Auto / chart toolbar	Plotly bar chart with zoom, pan and export controls for asset categories.
View Country Data	Expandable panel	Reveals the full ranked, sourced metric table for the selected nation.

Filters, Buttons & Interaction Design

Global Filter

A single, always-visible **Select Country** dropdown in the sidebar drives the entire dashboard state. It supports type-ahead search across all 190+ countries in the dataset and defaults to the first country alphabetically on load.

Standard Toolbar Controls

- **Share** — generates a shareable link to the current dashboard view.
- **Star** — bookmarks the app for quick access.
- **Edit (pencil)** — opens the underlying app code for authorised editors.
- **Refresh** — re-runs the data pipeline and clears cached results.
- **Overflow menu (■)** — settings, deployment and record/replay options.
- **Manage app** — footer control for deployment status and app administration.

Chart-Level Controls

Each Plotly visualisation (e.g. Military Asset Strength) exposes a hover toolbar with camera/download, zoom, pan, box- and lasso-select, autoscale and reset-axes actions, giving analysts fine-grained control without leaving the page.

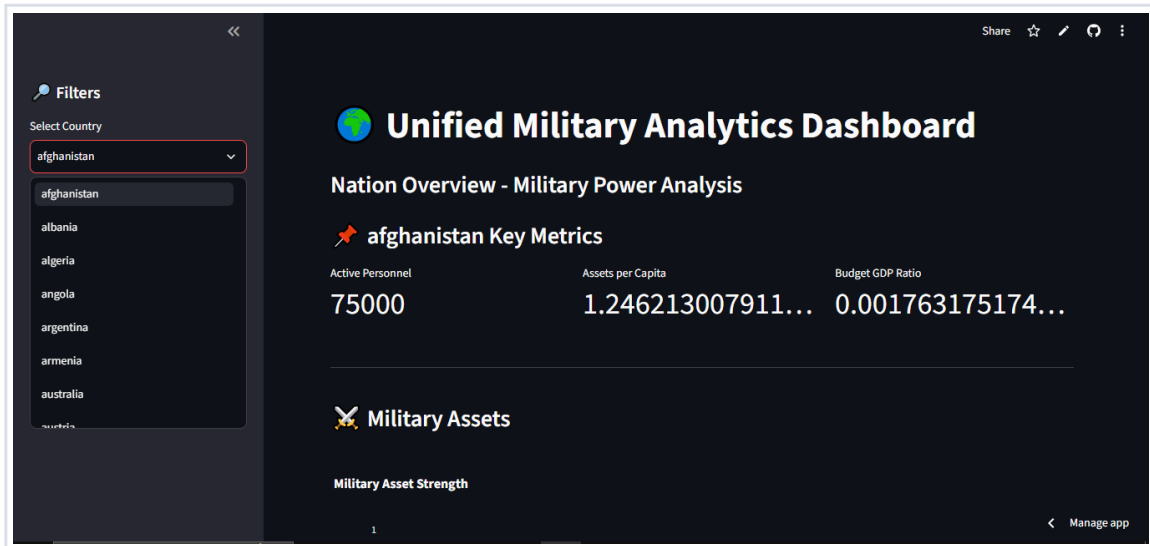
Data Table Interaction

The **View Country Data** panel is collapsible and presents every metric for the selected country as a ranked, sortable table with a direct **source URL** per row for full data traceability back to Global Firepower.

Storyboard — Nation Overview Module

The four frames below trace a single user journey: selecting a country, reviewing its headline KPIs, inspecting military asset strength, and drilling into the full sourced dataset.

1 Key Metrics — Country Selected



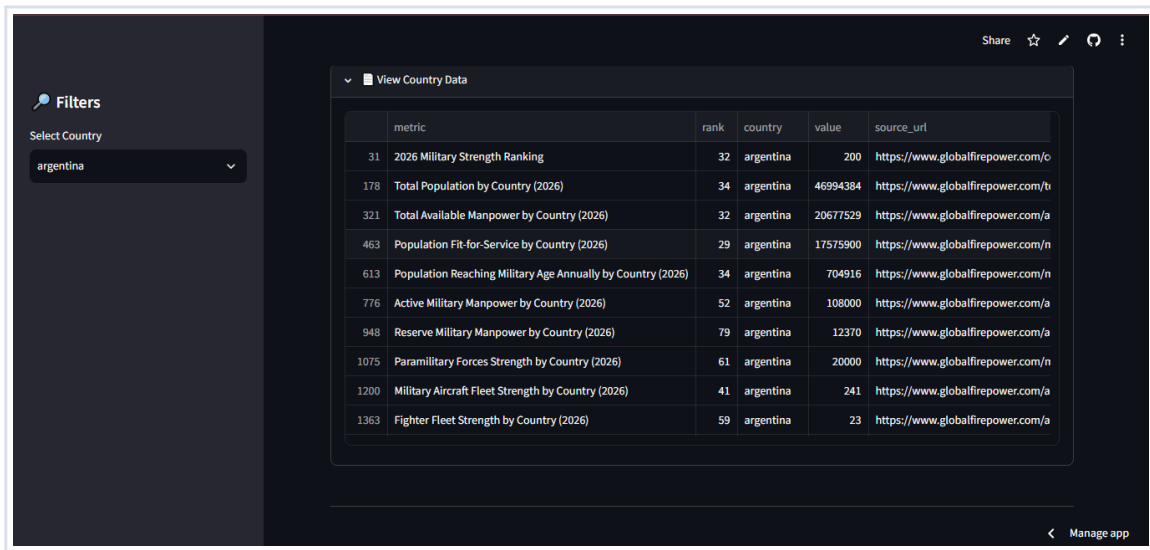
User selects a country from the sidebar filter (e.g. Afghanistan). The main canvas instantly renders three headline KPI cards: Active Personnel, Assets-per-Capita and Budget-to-GDP Ratio.

2 Military Assets — Asset Strength Chart



Scrolling down reveals the Military Asset Strength visualisation, an interactive Plotly bar chart (shown here for Tanks) with full zoom, pan and export tooling.

3 View Country Data — Ranked Metric Table



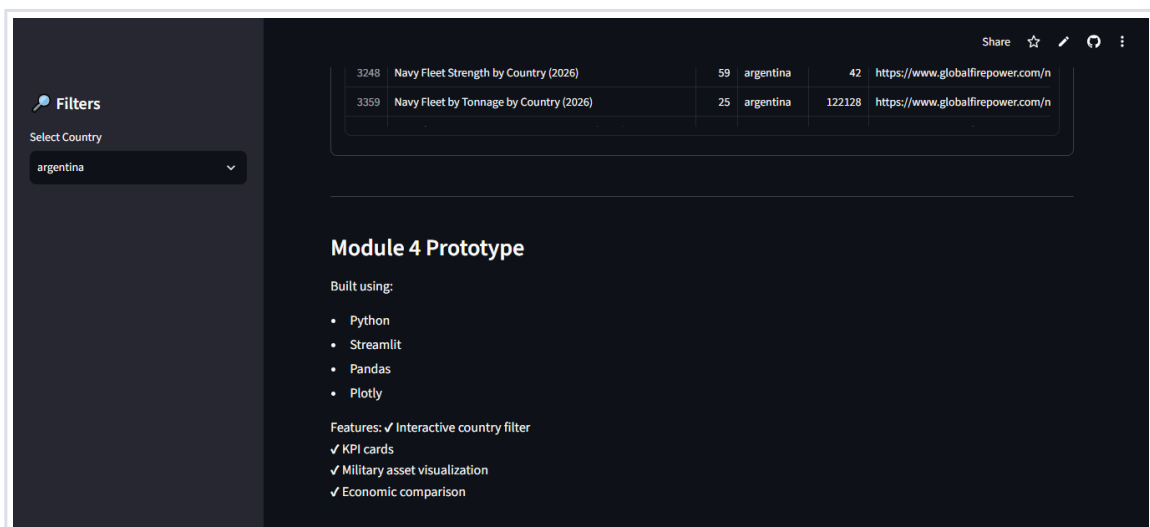
The screenshot shows a web application interface with a sidebar on the left containing a 'Filters' section with a 'Select Country' dropdown menu set to 'argentina'. The main content area is titled 'View Country Data' and displays a table of ranked metrics for Argentina. The table has columns for rank, country, value, and source_url. The data is as follows:

	metric	rank	country	value	source_url
31	2026 Military Strength Ranking	32	argentina	200	https://www.globalfirepower.com/c
178	Total Population by Country (2026)	34	argentina	46994384	https://www.globalfirepower.com/t
321	Total Available Manpower by Country (2026)	32	argentina	20677529	https://www.globalfirepower.com/a
463	Population Fit-for-Service by Country (2026)	29	argentina	17575900	https://www.globalfirepower.com/n
613	Population Reaching Military Age Annually by Country (2026)	34	argentina	704916	https://www.globalfirepower.com/n
776	Active Military Manpower by Country (2026)	52	argentina	108000	https://www.globalfirepower.com/a
948	Reserve Military Manpower by Country (2026)	79	argentina	12370	https://www.globalfirepower.com/a
1075	Paramilitary Forces Strength by Country (2026)	61	argentina	20000	https://www.globalfirepower.com/n
1200	Military Aircraft Fleet Strength by Country (2026)	41	argentina	241	https://www.globalfirepower.com/a
1363	Fighter Fleet Strength by Country (2026)	59	argentina	23	https://www.globalfirepower.com/a

At the bottom right of the interface, there is a 'Manage app' button.

The expandable data table lists every ranked metric for the country — population, manpower, reserve and paramilitary strength, aircraft and fighter fleets — each linked to its source.

4 Naval Metrics & Build Summary



The screenshot shows a web application interface with a sidebar on the left containing a 'Filters' section with a 'Select Country' dropdown menu set to 'argentina'. The main content area displays a table of naval metrics for Argentina. The table has columns for rank, country, value, and source_url. The data is as follows:

	metric	rank	country	value	source_url
3248	Navy Fleet Strength by Country (2026)	59	argentina	42	https://www.globalfirepower.com/n
3359	Navy Fleet by Tonnage by Country (2026)	25	argentina	122128	https://www.globalfirepower.com/n

Below the table, there is a section titled 'Module 4 Prototype' with the following content:

Built using:

- Python
- Streamlit
- Pandas
- Plotly

Features: ✓ Interactive country filter
✓ KPI cards
✓ Military asset visualization
✓ Economic comparison

The table continues into Navy Fleet Strength and Tonnage, followed by the in-app Module 4 build summary confirming the technology stack and completed feature set.

Application Framework	Streamlit	Renders the interactive web app, sidebar filters and layout.
Data Handling	Pandas · OpenPyXL	Loads, cleans and queries the military_final.xlsx dataset.
Visualisation	Plotly	Interactive, exportable charts for asset and comparison views.
Language	Python	Core application and data-processing logic.

Project Structure

```

Module_4/
  ■■ app.py
  ■■ requirements.txt
  ■■ military_final.xlsx
  ■■ dashboard_wireframe.pdf
  ■■ README.md
  
```


Deliverables & Evaluation Checklist

Deliverables

- ✓ Storyboard for dashboard layouts (this document, Section 04).
- ✓ A working dashboard application prototype, deployed with Streamlit.

Evaluation Criteria Coverage

Interactions and KPIs mapped correctly	Complete	Country filter drives KPI cards, chart and data table in real time.
One dashboard operational (Quick Stats / Nation Overview)	Complete	Nation Overview is fully functional end-to-end (Section 04).
Layout sketches / wireframes for all four views	Complete	Architecture and navigation logic documented in Section 02.
Filters, buttons and navigation logic defined	Complete	Detailed in Section 03.

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